

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 2.3 CHEMICAL BONDING: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use what you observed and learned in Lessons 1 to 8 to answer the driving question: Why do the aluminium sufuria, the graphite pencil and the table salt on our kitchen table look similar as solids but behave so differently, and what is happening between the atoms inside each one that controls every property we can measure? Write your answers in the blank space under each prompt. Use full sentences and particle-level words such as ions, lattice, delocalised electrons, covalent bonds and Van der Waals forces. For each claim, give the evidence from our tests and the reasoning that links the evidence to the particles. You may use a labelled sketch where a prompt allows it. Refer to your science journal and the Driving Question Board.

Part 1: Table Salt (Giant Ionic Structure)

The coastal table salt did not light the bulb as a dry solid, but it did light the bulb when dissolved in water. It also has a very high melting point of over 800 °C and shatters instead of bending. (a) Name the type of bond and the structure in solid NaCl and describe how it forms, using electrons. (b) Using a claim, evidence and reasoning (CER) structure, explain why dry salt does not conduct but salt solution does. (c) Explain why salt needs such a high temperature to melt. (d) Explain why a crystal of salt breaks instead of bending.

- (a) NaCl has ionic bonding in a giant ionic lattice. A sodium atom transfers its single valence electron to a chlorine atom. This makes Na^+ and Cl^- ions, each with a complete outer shell. The oppositely charged ions attract each other strongly and are arranged in a regular 3D lattice.
- (b) Claim: Dry salt does not conduct, but dissolved salt does. Evidence: The dry NaCl crystals did not light the bulb, distilled water did not light it, and NaCl dissolved in water did light it. Reasoning: In the solid, the Na^+ and Cl^- ions are locked in fixed positions and cannot move, and there are no free electrons, so nothing can carry charge. When salt dissolves in water (or melts), the lattice breaks apart and the ions become free to move to the electrodes. These mobile ions carry charge and complete the circuit.
- (c) Every ion is held by strong electrostatic forces acting in all directions throughout the lattice. A lot of energy is needed to overcome these forces, so the melting point is very high (over 800 °C, far beyond a home jiko).
- (d) A blow shifts one layer of ions so that ions of the same charge end up next to each other. Like charges repel, so the crystal splits instead of bending. This is what we saw in our maize-grain lattice model when one row was shifted.

Part 2: Aluminium (Giant Metallic Structure)

The aluminium wire from the sufuria handle was shiny, bent without breaking, and

- (a) Aluminium has metallic bonding in a giant metallic lattice. Al^{3+} cations sit in a regular lattice surrounded by a 'sea' of delocalised electrons that are free to move. The attraction between the cations and the electron sea holds the metal together.
- (b) In solid aluminium the delocalised electrons can move freely through the lattice, so they carry charge as soon as the circuit is

conducted electricity immediately as a solid. It melted at 660 °C, which is lower than salt. (a) Describe the bonding and structure inside aluminium. (b) Explain why aluminium conducts when solid, while solid NaCl does not. (c) Explain why aluminium bends instead of shattering. (d) Aluminium has strong bonding, yet it melts at a lower temperature than salt. What does this tell you about comparing the two structures?	connected. In solid NaCl the charged particles (ions) are locked in place and there are no free electrons, so no charge can flow. (c) The layers of cations can slide over each other while the electron sea keeps them bonded. Like charges are not forced together, so there is no repulsion that splits the metal. The metal is therefore malleable and ductile and bends without breaking. (d) Both structures are giant lattices with strong bonding, but the forces differ. Aluminium's melting point of 660 °C is lower than NaCl's, which is over 800 °C. So the bond type and structure decide the melting point, and we must compare the particular forces, not just say 'strong' or 'weak'. Here the ionic attractions in NaCl need more energy to overcome than the metallic bonding in aluminium.
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Part 3: Graphite (Giant Atomic / Layered Covalent Structure)	
The graphite rod from the AA battery is a non-metal made only of carbon, yet it conducts electricity as a solid. It is dull grey, snaps instead of bending, and leaves marks on paper. (a) Describe the bonding within a graphite layer and between layers. (b) Explain why graphite conducts even though carbon is a non-metal. (c) Explain why graphite leaves a mark on paper. (d) A classmate says 'only metals conduct electricity'. Use the evidence from our tests to respond.	(a) Within each layer, carbon atoms are covalently bonded in flat hexagons. Each carbon uses three of its four valence electrons in bonding. Between layers there are only weak Van der Waals forces. (b) Each carbon has one electron left over that is delocalised along the layer. These electrons are free to move and carry charge, so graphite conducts even though carbon is a non-metal. (c) The forces between layers are weak, so the layers slide apart easily. When a pencil is rubbed on paper, layers of graphite flake off and stay on the paper as a mark. (d) The rule 'only metals conduct' is not correct. Our tests showed that graphite, a non-metal, lit the bulb as a solid, while dry NaCl, which contains a metal, did not. Conduction depends on whether there are freely moving charged particles (delocalised electrons or mobile ions), not on whether the substance is a metal.

Part 4: Comparing Structures and Predicting New Substances				
(a) Complete this table in your answer by writing the structure type, the particles that are free to move (if any) in the solid, and one reason for each substance's melting point or mechanical behaviour: NaCl,	(a)			
	Substance	Structure type	Free to move in the solid	Reason for melting point or mechanical behaviour
	NaCl	Giant ionic	Nothing (ions are fixed)	Strong electrostatic forces throughout the lattice give a very high melting point (over

aluminium, graphite. (b) In Lesson 7 you predicted the properties of two new substances: iodine crystals (used in water-purification tablets sold in Nairobi) and iron in a Jua Kali welding rod. Choose ONE and predict its conductivity, melting point and mechanical behaviour, giving a reason from its structure.				800 °C); shifting layers causes repulsion so it is brittle
	Aluminium	Giant metallic	Delocalised electrons	Cation layers slide over the electron sea so it is malleable; melts at 660 °C
	Graphite	Giant atomic (layered covalent)	One delocalised electron per carbon, along each layer	Weak Van der Waals forces between layers let them slide, so graphite flakes and marks paper
(b) Iodine: Iodine is a simple molecular structure. Discrete I₂ molecules are held together by weak Van der Waals forces. It has no ions and no free electrons, so it does not conduct. It has a low melting point because little energy is needed to overcome the weak forces between molecules, and it is soft and easily broken. Alternatively, iron: Iron is a giant metallic structure with a sea of delocalised electrons. It conducts electricity as a solid, and its cation layers can slide so it is malleable and ductile. It has a high melting point because of strong metallic bonding.				

Part 5: The Complete Story and Revising Your Thinking

Write a final explanation of at least one paragraph that answers the driving question: Why do the sufuria aluminium, the graphite pencil and the table salt look similar as solids but behave so differently, and what is happening between the atoms that controls their properties? Your answer must (a) include all three substances, (b) link bond type to structure to properties, and (c) say how your thinking changed from your first prediction in Lesson 1. Then give one real-life reason for a material choice in Kenya based on what you learned.	The three kitchen items look similar as solids, but they have different bond types and therefore different structures, and these decide their properties. Table salt (NaCl) is a giant ionic lattice. Na⁺ and Cl⁻ ions are held by strong electrostatic attraction in fixed positions. So it does not conduct as a solid, but it conducts when dissolved or melted because the ions are free to move. It has a very high melting point (over 800 °C) and is brittle because shifting layers bring like charges together. Aluminium is a giant metallic structure with Al³⁺ cations in a sea of delocalised electrons. The electrons move freely so it conducts as a solid, and the layers of cations slide so it bends without breaking. Graphite is a giant atomic layered structure. Carbon atoms are covalently bonded in hexagonal layers, and each carbon has one delocalised electron that carries charge along the layer, so graphite conducts even though it is a non-metal. The layers are held by weak Van der Waals forces, so they slide apart and leave marks on paper. Bond type controls structure, and structure controls every property we can measure. In Lesson 1 I thought that only metals conduct and I could not explain salt or graphite. Now I know that conduction needs freely moving charged particles, and that depends on the bonding and structure. Real-life use: aluminium is used for sufurias because it conducts heat and can be shaped without breaking, and graphite is used for battery electrodes because it conducts electricity.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Linking bond type and structure to properties	Correctly identifies ionic, metallic and layered covalent bonding for all three substances and clearly links each structure to conductivity, melting point and mechanical behaviour.	Identifies the bond type for all three substances and links most properties to structure, with one or two gaps.	Names some bond types but describes properties without linking them to structure, or mixes up the substances.
Particle-level reasoning about conductivity	Explains conduction in terms of freely moving ions or delocalised electrons and states why they are free or fixed in each substance, including graphite and dry versus dissolved salt.	Explains conduction using mobile ions or electrons for most substances, but with minor omissions.	Relies on rules such as 'metals conduct' or 'ionic does not conduct' without describing particles.
Use of evidence (CER)	Supports every claim with specific evidence from our tests (for example the bulb lighting, 660 °C, over 800 °C, brittle versus bends) and clear reasoning.	Gives claims with some evidence from the tests, with reasoning that is sometimes incomplete.	Gives claims with little or no evidence, or evidence without reasoning.
Applying ideas to new substances and real life	Accurately predicts the properties of iodine or iron from its structure and gives a sound Kenyan real-life reason for a material choice.	Makes a mostly correct prediction with partial justification and a simple real-life link.	Prediction is incorrect or has no reasoning, and the real-life link is missing or unclear.
Scientific vocabulary and reflection	Uses terms such as lattice, electrostatic, delocalised and Van der Waals precisely, and reflects clearly on how thinking changed since Lesson 1.	Uses most terms correctly and gives a brief reflection on how thinking changed.	Uses few scientific terms correctly and gives little or no reflection.

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